MTH 101
Calculus & Analytical Geometry
Lessons 01–15

Virtual University of Pakistan

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1 Coordinates, Graphs and Lines

1.1 What is Calculus?

Calculus is the study of **continuous rates of change** of quantities.

1.2 Real Numbers

- **Natural Numbers:** $1, 2, 3, 4, \ldots$
- **Whole Numbers:** $0, 1, 2, 3, 4, \ldots$
- **Integers:** $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$
- **Rational Numbers:** $\frac{p}{q}$ where $q \neq 0$
- **Irrational Numbers:** Cannot be written as $\frac{p}{q}$ (e.g., $\sqrt{2}, \pi$)
- **Prime Numbers:** $2, 3, 5, 7, 11, 13, \ldots$

Hierarchy:

$$\text{Real} \begin{cases} \text{Rational} \begin{cases} \text{Integers} \begin{cases} \text{Whole} \begin{cases} \text{Natural} \end{cases} \end{cases} \end{cases} \\ \text{Irrational} \end{cases}$$

1.3 Sets

- A **set** is a collection of well-defined objects.
- Objects are called **elements**.
- $a \in A$ means a is an element of A.
- $a \notin A$ means a is not an element of A.
- $A \subseteq B$ means A is a subset of B.
- $\emptyset$ or $\{\}$ is the empty set.

Example:

$$A = \{1, 2, 3\}, \quad B = \{1, 2, 3, 4, 5\} \implies A \subseteq B$$

1.4 Union and Intersection

- $A \cup B$ = all elements in A **or** B.
- $A \cap B$ = all elements in **both** A and B.

Example:

$$A \cup B = \{1, 2, 3, 4, 5\}, \quad A \cap B = \{1, 2, 3\}$$

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1.5 Intervals

Notation	Definition	Type
$[a, b]$	$\{x : a \leq x \leq b\}$	Closed
(a, b)	$\{x : a < x < b\}$	Open
$[a, b)$	$\{x : a \leq x < b\}$	Half-open
$(a, b]$	$\{x : a < x \leq b\}$	Half-open
$(-\infty, b]$	$\{x : x \leq b\}$	Closed
$(-\infty, b)$	$\{x : x < b\}$	Open

1.6 Solving Inequalities

Example 1: Solve $3 + 7x \leq 2x - 9$.

$$\begin{aligned} 7x - 2x &\leq -9 - 3 \\ 5x &\leq -12 \\ x &\leq -\frac{12}{5} \end{aligned}$$

Solution: $\left(-\infty, -\frac{12}{5}\right]$.

Example 2: Solve $7 \leq 2 - 5x < 9$.

$$\begin{aligned} 5 &\leq -5x < 7 \\ -1 &\geq x > -\frac{7}{5} \\ -\frac{7}{5} &< x \leq -1 \end{aligned}$$

Solution: $\left(-\frac{7}{5}, -1\right]$.

2 Absolute Value

2.1 Definition

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

Examples: $|5| = 5$, $\left|-\frac{4}{7}\right| = \frac{4}{7}$, $|0| = 0$.

2.2 Properties

1. $|-a| = |a|$
2. $|ab| = |a||b|$
3. $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$
4. $\sqrt{a^2} = |a|$

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2.3 Equations with Absolute Value

Example 1: $|x-3|=4$

$$x-3=4 \quad \text{or} \quad x-3=-4 \implies x=7 \text{ or } x=-1$$

Example 2: $|3x-2|=|5x+4|$

$$3x-2=5x+4 \implies x=-3$$

$$3x-2=-(5x+4) \implies x=-\frac{1}{4}$$

2.4 Inequalities with Absolute Value

Example 1: $|x-3|<4$

$$-4<x-3<4 \implies -1<x<7$$

Solution: $(-1,7)$.

Example 2: $|x+4|\geq 2$

$$x+4\leq -2 \quad \text{or} \quad x+4\geq 2$$

$$x\leq -6 \quad \text{or} \quad x\geq -2$$

Solution: $(-\infty,-6]\cup[-2,+\infty)$.

2.5 Distance Formula

$$d=|b-a|$$

Geometric meaning:

- $|x-a|$ = distance between x and a
- $|x+a|$ = distance between x and $-a$
- $|x|$ = distance between x and origin

2.6 Triangle Inequality

$$|a+b|\leq |a|+|b|$$

3 Coordinate Planes and Graphs

3.1 The Coordinate Plane

A plane is the intersection of two coordinate lines at 90°, called the **coordinate plane** or xy**-plane**.

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3.2 Quadrants

Quadrant	Signs
I	$(+,+)$
II	$(-,+)$
III	$(-,-)$
IV	$(+,-)$

3.3 Intercepts

- **x-intercept:** Set $y = 0$, solve for x.
- **y-intercept:** Set $x = 0$, solve for y.

Example 1: $3x + 2y = 6$

- x-intercept: $3x = 6 \implies x = 2$
- y-intercept: $2y = 6 \implies y = 3$

Example 2: $x = y^2 - 2y$

- x-intercept: $x = 0$
- y-intercept: $0 = y(y-2) \implies y = 0, 2$

3.4 Symmetry

Points	Symmetric about
(x,y) and $(-x,y)$	y-axis
(x,y) and $(x,-y)$	x-axis
(x,y) and $(-x,-y)$	Origin

Example 1: $y = \frac{1}{8}x^4 - x^2$
Replace x by $-x$:

$$y = \frac{1}{8}(-x)^4 - (-x)^2 = \frac{1}{8}x^4 - x^2$$

Unchanged $\implies$ symmetric about y-axis.

Example 2: $x = y^2$
Replace y by $-y$:

$$x = (-y)^2 = y^2$$

Unchanged $\implies$ symmetric about x-axis.

4 Lines and Slope

4.1 Slope

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

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Example 1: Through $(6, 2)$ and $(9, 8)$:

$$m = \frac{8-2}{9-6} = \frac{6}{3} = 2$$

Example 2: Through $(-2, 7)$ and $(5, 7)$:

$$m = \frac{7-7}{5-(-2)} = \frac{0}{7} = 0$$

4.2 Interpretation of Slope

Slope	Meaning
Positive	Inclined upward to the right
Negative	Inclined downward to the right
Zero	Horizontal
Undefined	Vertical

4.3 Angle of Inclination

For a non-vertical line:

$$m = \tan\phi$$

where $0° \leq \phi \leq 180°$.

Example 1: $m = 1$

$$1 = \tan\phi \implies \phi = 45°$$

Example 2: $m = -1$

$$-1 = \tan\phi \implies \phi = 135°$$

4.4 Parallel and Perpendicular Lines

- Parallel: $m_1 = m_2$
- Perpendicular: $m_1 \cdot m_2 = -1$

Example: Show $A(1, 3)$, $B(3, 7)$, $C(7, 5)$ form a right triangle.

$$m_{AB} = \frac{7-3}{3-1} = 2, \quad m_{BC} = \frac{5-7}{7-3} = -\frac{1}{2}$$

$$m_{AB} \cdot m_{BC} = 2 \cdot \left(-\frac{1}{2}\right) = -1$$

$\implies$ Right triangle.

4.5 Equations of Lines

Form	Equation
Vertical	$x = a$
Horizontal	$y = b$
Point-slope	$y - y_1 = m(x - x_1)$
Slope-intercept	$y = mx + b$
General	$Ax + By + C = 0$

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Example 1: Line through $(2, 3)$ with slope $-\frac{3}{2}$:

$$y - 3 = -\frac{3}{2}(x - 2) \implies y = -\frac{3}{2}x + 6$$

Example 2: Line through $(-2, -1)$ and $(3, 4)$:

$$m = \frac{4 - (-1)}{3 - (-2)} = 1$$

$$y - 4 = 1(x - 3) \implies y = x + 1$$

Example 3: $8x + 5y = 20$:

$$y = -\frac{8}{5}x + 4 \implies m = -\frac{8}{5}, \quad b = 4$$

5 Distance, Circles, Quadratic Equations

5.1 Distance Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example 1: Distance between $(-2, 3)$ and $(1, 7)$:

$$d = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

Example 2: Show $A(4, 6)$, $B(1, -3)$, $C(7, 5)$ form right triangle:

$$AB = \sqrt{90}, \quad AC = \sqrt{10}, \quad BC = 10$$

$$AB^2 + AC^2 = 90 + 10 = 100 = BC^2$$

5.2 Midpoint Formula

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Example: Midpoint of $(3, -4)$ and $(7, 2)$:

$$(5, -1)$$

5.3 Circles

Standard form: $(x - x_0)^2 + (y - y_0)^2 = r^2$

Example 1: Circle radius 4 centered at $(-5, 3)$:

$$(x + 5)^2 + (y - 3)^2 = 16$$

Example 2: Circle center $(1, -2)$ through $(4, 2)$:

$$r = \sqrt{9 + 16} = 5$$

$$(x - 1)^2 + (y + 2)^2 = 25$$

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General form: $x^2 + y^2 + Dx + Ey + F = 0$
Example: Find center and radius of $x^2 + y^2 - 8x + 2y + 8 = 0$:

$$(x-4)^2 + (y+1)^2 = 9$$

Center: $(4, -1)$, Radius: 3.
Degenerate cases:

Condition	Graph
$k > 0$	Circle
$k = 0$	Single point
$k < 0$	No graph

5.4 Quadratic Equations

Form: $y = ax^2 + bx + c$ $(a \neq 0)$
Vertex: $x = -\frac{b}{2a}$
Example: $y = -x^2 + 4x - 5$:

$$x = 2, \quad y = -1 \implies (2, -1)$$

Example: Ball height $s = 24.5t - 4.9t^2$:

$$t = 2.5 \text{ sec}, \quad s = 30.625 \text{ m}$$

6 Functions

6.1 Definition

A function assigns to each x exactly one y.
Notation: $y = f(x)$
Example: $f(x) = x^2 \implies f(3) = 9,\ f(-2) = 4.$
Not a function: $y = \pm\sqrt{x}$.

6.2 Domain

- **Natural domain:** Determined by formula.
- **Restricted domain:** Altered by algebraic manipulation.

Example: $h(x) = \frac{1}{(x-1)(x-3)}$ Domain: $(-\infty, 1) \cup (1, 3) \cup (3, +\infty)$.
Example: $h(x) = \frac{x^2-4}{x-2} = x + 2, \quad x \neq 2$
Example: Square cut from 10 cm cardboard:

$$y = 100 - 4x^2, \quad 0 \leq x \leq 5$$

6.3 Range

Example: $f(x) = x^2 \implies$ Range: $[0, +\infty)$.
Example: $y = \frac{x+1}{x-1}$:

$$x = \frac{y+1}{y-1} \implies y \neq 1$$

Range: $(-\infty, 1) \cup (1, +\infty)$.

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6.4 Piecewise Functions

Example: Taxicab fare:

$$f(x) = \begin{cases} 1.75 & 0 < x \leq 1 \\ 1.75 + 0.50(x-1) & x > 1 \end{cases}$$

6.5 Operations on Functions

$$(f+g)(x) = f(x) + g(x)$$
$$(f-g)(x) = f(x) - g(x)$$
$$(f \cdot g)(x) = f(x) \cdot g(x)$$
$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \quad g(x) \neq 0$$

6.6 Composition of Functions

$$(f \circ g)(x) = f(g(x))$$

Example: $f(x) = x^2 + 3$, $g(x) = \sqrt{x}$:

$$(f \circ g)(x) = (\sqrt{x})^2 + 3 = x + 3$$

Domain: $[0, +\infty)$.

6.7 Classification of Functions

Type	Form
Constant	$f(x) = c$
Monomial	cx^n (n non-negative integer)
Polynomial	$a_0 + a_1 x + \cdots + a_n x^n$

7 Graphs of Functions

7.1 Graph of a Function

Graph of f = graph of $y = f(x)$.

Example: $f(x) = |x|$ (V-shaped).

Example: $f(x) = \frac{x^2-4}{x-2} = x + 2, \quad x \neq 2$ (line with hole).

7.2 Translations

Transformation	Effect
$y = f(x) + c$	Shift up by c
$y = f(x) - c$	Shift down by c
$y = f(x + c)$	Shift left by c
$y = f(x - c)$	Shift right by c

Example: $y = |x - 3| + 2 \implies$ shift right 3, up 2.

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7.3 Reflections

Transformation	Effect
$y = f(-x)$	Reflection about y-axis
$y = -f(x)$	Reflection about x-axis

7.4 Vertical Line Test

A graph is a function iff no vertical line intersects it more than once.

Example: $x^2 + y^2 = 25$ (circle) $\implies$ not a function.

8 Limits (Intuitive Introduction)

8.1 Definition

$$\lim_{x \to a} f(x) = L$$

means as x approaches a, $f(x)$ approaches L.

8.2 Left and Right Limits

$$\lim_{x \to a^-} f(x) = L_1, \quad \lim_{x \to a^+} f(x) = L_2$$

Limit exists iff $L_1 = L_2$.

Example: $\lim_{x \to 0} \frac{\sin x}{x} = 1$

8.3 Existence of Limits

Limits fail due to:

1. Oscillation (e.g., $\sin(1/x)$ as $x \to 0$)
2. Unbounded increase/decrease (e.g., $1/x$ as $x \to 0$)

9 Limits — Computational Techniques

9.1 Basic Limits

$$\lim_{x \to a} k = k, \quad \lim_{x \to a} x = a, \quad \lim_{x \to a} x^n = a^n$$

9.2 Properties

$$\lim_{x \to a}[f(x) + g(x)] = \lim f(x) + \lim g(x)$$
$$\lim_{x \to a}[f(x) - g(x)] = \lim f(x) - \lim g(x)$$
$$\lim_{x \to a}[f(x) \cdot g(x)] = \lim f(x) \cdot \lim g(x)$$
$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim f(x)}{\lim g(x)}, \quad \lim g(x) \neq 0$$

Example: $\lim_{x \to 5}(x^2 - 4x + 3) = 25 - 20 + 3 = 8$.

Example: $\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = \lim_{x \to 2}(x + 2) = 4$.

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9.3 Limits Involving $\frac{1}{x}$

$$\lim_{x\to 0^+}\frac{1}{x}=+\infty,\quad \lim_{x\to 0^-}\frac{1}{x}=-\infty$$

$$\lim_{x\to +\infty}\frac{1}{x}=0,\quad \lim_{x\to -\infty}\frac{1}{x}=0$$

9.4 Quick Rule for Rational Functions

$$\lim_{x\to\pm\infty}\frac{c_0+c_1x+\cdots+c_nx^n}{d_0+d_1x+\cdots+d_mx^m}=\lim_{x\to\pm\infty}\frac{c_nx^n}{d_mx^m}$$

Example: $\lim_{x\to-\infty}\frac{4x^2-x}{2x^3-5}=0$.

10 Limits — Rigorous Approach

10.1 ϵ-δ Definition

$$\lim_{x\to a}f(x)=L$$

means: For every $\epsilon>0$, there exists $\delta>0$ such that if $0<|x-a|<\delta$, then $|f(x)-L|<\epsilon$.

Example: Prove $\lim_{x\to 2}(3x-5)=1$.

$$|(3x-5)-1|<\epsilon \implies 3|x-2|<\epsilon \implies |x-2|<\frac{\epsilon}{3}$$

Choose $\delta=\frac{\epsilon}{3}$.

11 Continuity

11.1 Definition

f is **continuous** at c if:

1. $f(c)$ is defined.
2. $\lim_{x\to c}f(x)$ exists.
3. $\lim_{x\to c}f(x)=f(c)$.

11.2 Types of Discontinuity

Type	Description
Removable	Limit exists but $f(c)$ undefined/different
Jump	Left and right limits differ
Infinite	Limit is infinite

Example: $f(x)=\frac{x^2-4}{x-2}$ discontinuous at $x=2$.

Example: $g(x)=\begin{cases}\frac{x^2-4}{x-2} & x\neq 2\\ 3 & x=2\end{cases}$

$\lim_{x\to 2}g(x)=4\neq g(2)=3 \implies$ discontinuous.

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11.3 Properties

If f, g continuous at c:

- $f+g$, $f-g$, $f \cdot g$ continuous at c.
- $\frac{f}{g}$ continuous at c if $g(c) \neq 0$.

11.4 Polynomials and Rational Functions

- Polynomials continuous everywhere.
- Rational functions continuous except where denominator $= 0$.

11.5 Intermediate Value Theorem

If f continuous on $[a, b]$ and C between $f(a)$ and $f(b)$, then exists $x \in [a, b]$ with $f(x) = C$.

12 Limits and Continuity of Trigonometric Functions

12.1 Continuity of Sine and Cosine

$$\lim_{x \to 0} \sin x = 0, \quad \lim_{x \to 0} \cos x = 1$$

Both continuous everywhere.

12.2 Other Trigonometric Functions

Function	Continuous except at
$\tan x$	$x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
$\cot x$	$x = 0, \pm\pi, \pm 2\pi, \ldots$
$\sec x$	$x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
$\csc x$	$x = 0, \pm\pi, \pm 2\pi, \ldots$

12.3 Squeeze Theorem

If $g(x) \leq f(x) \leq h(x)$ near a and $\lim g = \lim h = L$, then $\lim f = L$.

12.4 Important Limits

$$\lim_{x \to 0} \frac{\sin x}{x} = 1, \quad \lim_{x \to 0} \frac{1 - \cos x}{x} = 0$$

Proof of first limit: Using areas in unit circle:

$$0 < \frac{1}{2} \sin x < \frac{1}{2} x < \frac{1}{2} \tan x$$

$$\sin x < x < \tan x \implies \cos x < \frac{\sin x}{x} < 1$$

As $x \to 0$, $\cos x \to 1 \implies$ by Squeeze Theorem, $\frac{\sin x}{x} \to 1$.

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13 Tangent Lines and Rates of Change

13.1 Secant and Tangent Lines

$$m_{\text{sec}} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$m_{\tan} = \lim_{x_1 \to x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

13.2 Average and Instantaneous Velocity

$$v_{\text{ave}} = \frac{f(t_1) - f(t_0)}{t_1 - t_0}$$

$$v_{\text{inst}} = \lim_{t_1 \to t_0} \frac{f(t_1) - f(t_0)}{t_1 - t_0}$$

13.3 Average and Instantaneous Rate of Change

- **Average:** Slope of secant line.
- **Instantaneous:** Slope of tangent line.

Example: $y = f(x) = x^2 + 1$
(a) Average rate over $[3, 5]$**:**

$$m_{\text{sec}} = \frac{26 - 10}{2} = 8$$

(b) Instantaneous rate at $x_0 = -4$**:**

$$m_{\tan} = \lim_{x_1 \to -4} \frac{x_1^2 - 16}{x_1 + 4} = \lim_{x_1 \to -4} (x_1 - 4) = -8$$

(c) General:

$$m_{\tan} = \lim_{x_1 \to x_0} \frac{x_1^2 - x_0^2}{x_1 - x_0} = 2x_0$$

14 The Derivative

14.1 Definition

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Notation: $f'(x) = \frac{dy}{dx} = \frac{d}{dx}[f(x)]$

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14.2 Tangent Line Equation

$$y - y_0 = f'(x_0)(x - x_0)$$

Example: Tangent to $f(x) = x^2$ at $(3, 9)$:

$$f'(3) = \lim_{h\to 0} \frac{(3+h)^2 - 9}{h} = \lim_{h\to 0}(6+h) = 6$$

$$y - 9 = 6(x - 3) \implies y = 6x - 9$$

Example: $f(x) = x^2 + 1$:

$$f'(x) = \lim_{h\to 0} \frac{(x+h)^2 + 1 - (x^2+1)}{h} = \lim_{h\to 0}(2x+h) = 2x$$

Example: $f(x) = \sqrt{x}$:

$$f'(x) = \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{1}{2\sqrt{x}}$$

14.3 Differentiability and Continuity

If f differentiable at x_0, then f continuous at x_0.

Non-differentiability occurs at:

1. Corners
2. Vertical tangents
3. Points of discontinuity

15 Techniques of Differentiation

15.1 Constant Function

$$\frac{d}{dx}[c] = 0$$

15.2 Power Rule

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

Example: $\frac{d}{dx}[x^5] = 5x^4$

15.3 Constant Multiple Rule

$$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}[f(x)]$$

Example: $\frac{d}{dx}[3x^8] = 24x^7$

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15.4 Sum and Difference Rules

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}[f(x)] \pm \frac{d}{dx}[g(x)]$$

Example: $\frac{d}{dx}[x^4 + x^3] = 4x^3 + 3x^2$

15.5 Product Rule

$$\frac{d}{dx}[f(x)g(x)] = f(x)\frac{d}{dx}[g(x)] + g(x)\frac{d}{dx}[f(x)]$$

15.6 Quotient Rule

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}[f(x)] - f(x)\frac{d}{dx}[g(x)]}{[g(x)]^2}$$

15.7 Reciprocal Rule

$$\frac{d}{dx}\left[\frac{1}{g(x)}\right] = -\frac{\frac{d}{dx}[g(x)]}{[g(x)]^2}$$

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